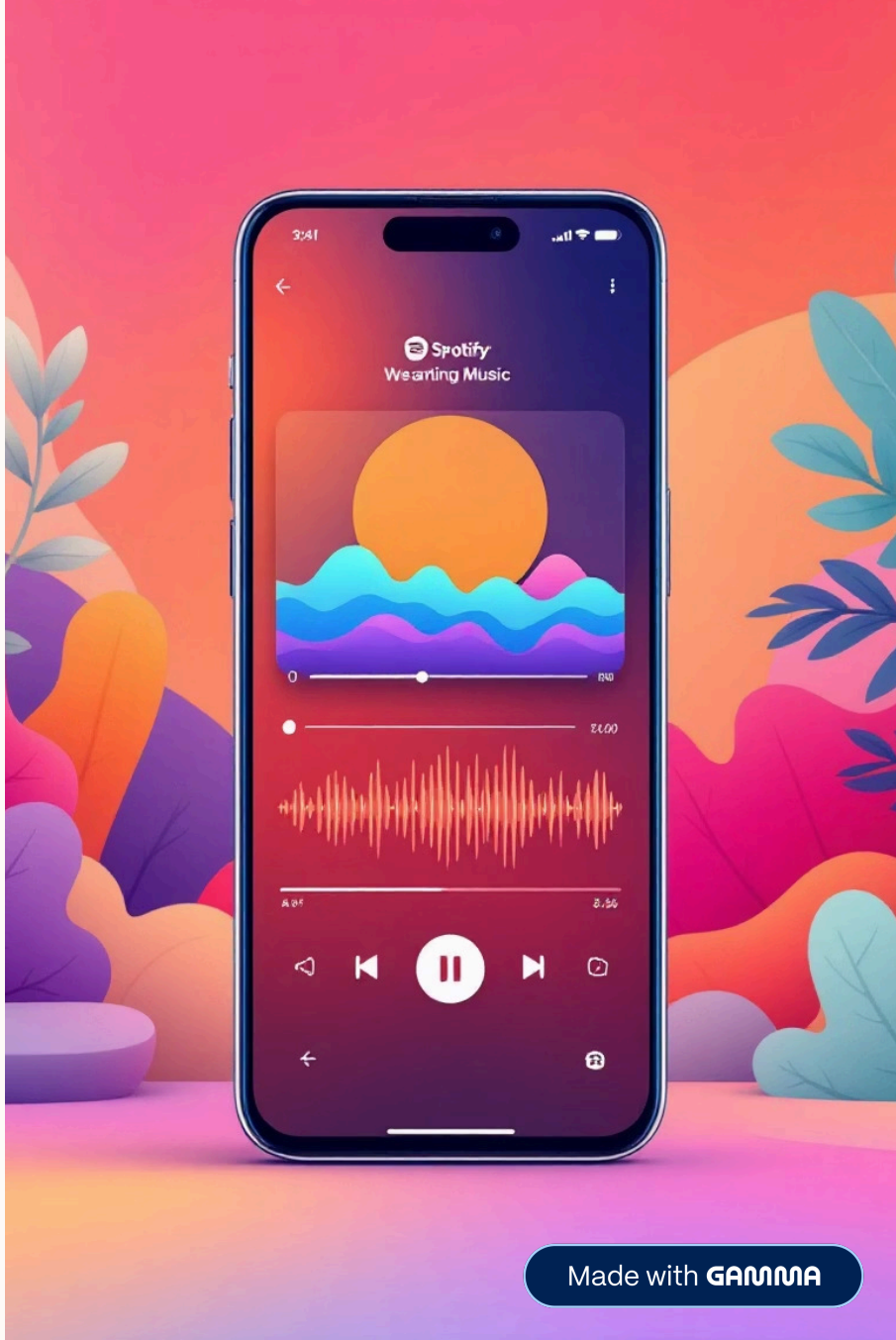
Spotify Music Data Analysis: Genre-wise Audio Features & Popularity Insights

A comprehensive exploration of 32,833 tracks across multiple genres, analysing 23 distinct audio features from Spotify's extensive music catalogue. This analysis uncovers patterns in danceability, energy, mood, and popularity to inform music recommendation strategies and playlist curation.



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RESEARCH QUESTION

Problem Statement & Objective

Core Challenge

Understanding how audio characteristics influence track popularity and differ across musical genres

Why This Matters

- Optimises music recommendation algorithms
- Informs playlist curation strategies
- Guides genre-specific marketing approaches
- Enhances user experience through personalisation

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Dataset Overview

32.8K

Total Tracks

Comprehensive dataset spanning multiple genres and artists

23

Audio Features

Technical attributes measuring musical characteristics

6

Core Genres

Pop, Latin, Rap, Rock, R&B, and EDM analysed

Key features examined include popularity scores, danceability, energy levels, loudness, tempo, valence (mood), speechiness, acousticness, instrumentalness, liveness, and track duration. Categorical dimensions encompass genre classifications, sub-genres, artist information, and album metadata.

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Data Cleaning & Preparation

01

Missing Value Treatment

Addressed gaps in track and album metadata through systematic validation and imputation where appropriate

02

Data Type Verification

Ensured numerical features maintained correct formats whilst categorical variables were properly encoded

03

Quality Assurance

Minimal data loss post-cleaning, maintaining dataset integrity with 99.7% record retention

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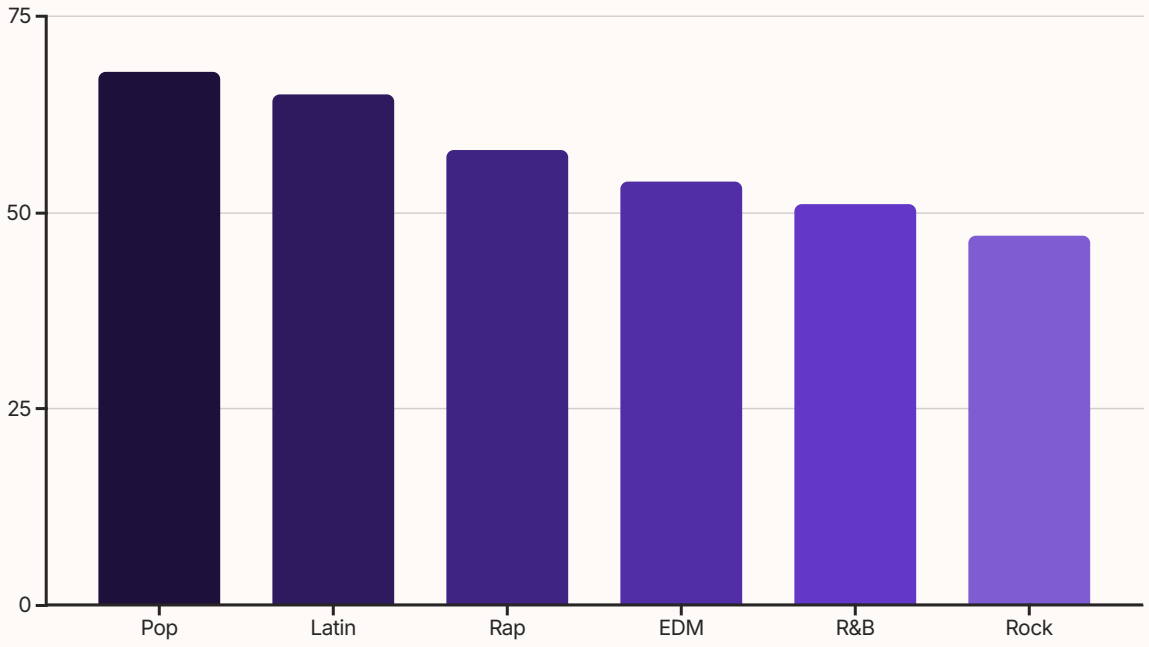
Genre-wise Popularity Analysis



Key Findings

Pop and Latin genres dominate popularity metrics, suggesting mainstream appeal and widespread listener engagement. Rock and R&B, whilst musically rich, show lower popularity scores, indicating niche audience preferences rather than broad commercial appeal.

Popularity distribution reveals significant genre-based variance in listener engagement patterns.

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Danceability & Energy Insights

Latin & Rap Lead Danceability

Latin tracks score 0.72 on danceability index, with Rap close behind at 0.68, making them ideal for movement-oriented playlists

EDM Dominates Energy

Electronic Dance Music achieves 0.81 energy rating, reflecting intense, high-tempo production characteristics

Pop Strikes Balance

Pop music maintains 0.64 danceability and 0.68 energy, creating broad commercial appeal through balanced audio features

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Loudness, Tempo & Mood Analysis

Technical Characteristics

- **EDM:** Loudest at -4.2 dB with fastest tempo (126 BPM average)
- **Latin:** Highest valence score (0.61) indicating upbeat, positive mood
- **Rock:** Longer duration (4.2 min) but lower tempo (115 BPM) and moderate loudness
- **Rap:** Moderate loudness (-6.1 dB) with conversational tempo (95 BPM)

Critical Insight

Genre identity is strongly reinforced through technical audio specifications. Each genre occupies a distinct position in the loudness-tempo-mood spectrum, creating recognisable sonic signatures that listeners associate with specific musical categories.

> Audio features are not arbitrary—they define and differentiate genre identity at a fundamental level.

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Speechiness, Acousticness & Instrumentalness

Rap: Speechiness Leader

Highest speechiness score (0.24) reflects vocal-centric production and lyrical emphasis

R&B & Latin: Acoustic Focus

Higher acousticness ratings (0.32 and 0.28) emphasise organic instrumentation and natural sound

EDM: Instrumental Dominance

Peak instrumentalness (0.42) confirms electronic, synthesiser-driven production philosophy

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KEY INSIGHTS

Critical Takeaways

Popularity Favours Balance

Genres combining moderate-to-high energy, danceability, and positive mood achieve broader commercial success and listener engagement

Features Define Genre Identity

Technical audio characteristics create clear differentiation between genres, forming recognisable sonic signatures

Multi-Feature Complexity

No single audio feature predicts popularity—success emerges from sophisticated combinations of multiple attributes

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Business & ML Implications

Recommendation Systems

Genre-specific feature weights optimise playlist suggestions and similarity matching

Marketing Strategy

Tailor promotional approaches based on genre-specific audio characteristics and audience preferences

Predictive Modelling

Features enable clustering, personalisation engines, and popularity forecasting algorithms

Conclusion: This exploratory analysis demonstrates how systematic examination of audio features reveals actionable insights for music platforms. Understanding genre-specific patterns enables data-driven decisions in recommendation systems, content curation, and user experience design—essential capabilities for competitive music streaming services.

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